

RelCompat3D: Re-Ranking 3D Scene Graph Relations with Geometric Evidence

Supplementary Material

Anonymous submission

A Supplementary Method Details

This section provides the method details and formal properties.

A.1 Notation

Table S1 summarizes the notation used in the main paper and the supplement.

Symbol	Meaning
r_i	Relation candidate i
(s_i, o_i)	Ordered subject and object instance identifiers
p_i, a_i	Predicate label and relation-family label
$k_i^{\text{pair}}, k_i^{\text{rel}}$	Ordered-pair identity and exact relation-candidate identity
T_i, G_i, Z_i	Predicate semantics, ordered-pair measurements, and source relation score
q, f_q, C_i^q	Compatibility estimator, compatibility logit, and bounded compatibility score
$H_{a_i}, \mathcal{O}_i$	Applicable transformation group and transformation orbit
$C_i^{\text{tr},q}, u_i^q$	Transformation-consistent compatibility and within-family ranking score
$\mathcal{A}_{\text{eval}}, \mathcal{A}_{\text{rank}}$	Evaluated and re-ranked relation-family sets
π^Z, π_a^q	Source ranking and ordered list for family a
$L_K(c), Y_c$	Selected top- K candidates and exact ground-truth relations in context c
D_K, N_s, N_u, N_v	Status-assigned candidates and satisfied, uncertain, and violated counts
$\mathcal{P}, \ell_i^q$	Linked training pairs and compatibility logit used by the pairwise loss

Table S1: Notation used in the RelCompat3D method and evaluation.

A.2 Counterfactual Construction and Information Use

Ground-truth relations in the evaluated families provide positives. Negatives are generated within the same scene context and linked to their source positive. The construction does not

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use source relation scores, predictor identities, or evaluation candidates. Table S2 gives the complete construction policy.

Separation of training and evaluation. Table S3 shows which information each procedure uses. Training examples come only from the training split, so evaluation candidate rows, source relation scores, and primary verifier-status labels do not enter counterfactual construction. The counterfactual rules and primary verifier still share some OBB-derived measurements and family thresholds. The point- and mesh-based audit does not use those OBB measurements, although it uses the same reconstructed evaluation scenes and relation ontology.

We limit negatives to two per positive, 200 for each combination of context and family, and three times the number of positives per family. These limits reduce repeated easy negatives without equalizing family counts. The constructed training and development sets together contain 28,977 positives and 37,477 negatives. Their combined family totals are 23,008 proximity, 5,488 vertical, and 37,958 support/contact rows. Because training construction and the verifier use some of the same measurements, Violation is reported as a verifier-derived measure rather than independent ground truth for geometric validity. The feature-removal analysis tests this dependence but does not eliminate it.

A.3 Estimator Architecture and Optimization

RelCompat3D-Linear fits one logistic head for each evaluated relation family. Numeric features use means and standard deviations estimated from the training split. Missing values use the corresponding training mean. Inputs are predicate indicators T , predicate-independent ordered-pair measurements G , and the two signed-height interactions defined in the main paper. The relation-family label selects the head and is not an input feature. The source relation score, source rank, predictor identity, and object class are excluded. Relation-preserving augmentation keeps the proximity predicate fixed under endpoint swap and applies the inverse predicate after a vertical-order endpoint swap. Training applies the margin loss from the main paper to linked positive-counterfactual pairs. Support/contact receives the pairwise loss and uses only the identity transformation.

Geometric feature specification. Table S4 defines the 17 measurements in G . For endpoint $j \in \{s, o\}$, let $c_j =$

Family	Construction rule	Relation to verifier
Proximity	Choose ordered pairs from the same context in descending normalized XY distance. Require normalized distance ≥ 2.5 and zero projected overlap.	Uses the same distance and overlap measurements. With zero overlap, the verifier labels distance ≤ 2.5 satisfied and distance ≥ 3.5 violated.
Vertical	On the same ordered pair, replace <code>higher than</code> with <code>lower than</code> or conversely. Require the positive direction to have $ \Delta z \geq 0.25$ m and normalized $ \Delta z \geq 0.15$.	Uses the same signed measurements and thresholds.
Support/contact	Replace the subject or object within the context. Reject known support/contact pairs in either direction and pairs whose subject is the floor. Require any two of normalized XY distance ≥ 2.0 , zero overlap, and vertical gap ≥ 0.30 m.	Uses the same OBB measurements and thresholds. Point evidence used by the verifier for each support type is not used to generate negatives.

Table S2: Counterfactual-negative construction. Normalized XY distance uses the mean ordered-pair XY diagonal. Zero overlap requires both directional projected-overlap ratios to be at most 10^{-9} .

Information	Train	Verifier	Audit
Evaluation candidates	No	Yes	Yes
Source relation score	No	No	No
Verifier labels	No	Output	No
OBB measurements	Some	Yes	No
Point- and mesh-based measurements	No	No	Yes
Evaluation scene identities	No	Yes	Yes
Relation ontology	Yes	Yes	Yes

Table S3: Information used by counterfactual training construction (Train), the primary verifier (Verifier), and the point- and mesh-based audit (Audit). “Some” denotes the OBB measurements and family thresholds shared by training construction and the primary verifier. The audit uses alternative measurements rather than independent ground truth for geometric validity.

(x_j, y_j, z_j) be the OBB center and b_j and t_j its lower and upper vertical bounds. We set $h_j = t_j - b_j$, and d_j^{3D} and d_j^{XY} denote the diagonals of its 3D and projected XY bounds. Let A_j denote the area of the projected bounds and I_{XY} their intersection area. All distance and height measurements are in meters. Normalized and overlap measurements are dimensionless.

RelCompat3D-MLP fits one shared multilayer perceptron (MLP) with a single two-unit ReLU hidden layer and a predicate-linear skip connection. Its 29 inputs include relation-family and predicate indicators, the same 17 geometric measurements, the two signed-height interactions, and the sum of the directional overlap ratios. Because parameters are shared across families, the relation-family indicator varies across inputs. The estimator has 69 parameters and excludes the source relation score and predictor identity.

The 60,208 training rows yield 33,961 linked pairs, and relation-preserving augmentation produces 86,032 optimization rows for both estimators. Linear uses 800 deterministic full-batch gradient descent steps at learning rate 0.2. MLP uses 120 full-batch Adam steps at learning rate 0.02. Both use the same pairwise weight, margin, and ℓ_2 coefficient. The three linear family models contain 21–23 parameters. Each reported condition is fitted once with a pre-specified

seed. The paired intervals quantify scan resampling rather than retraining variability.

A.4 Formal Properties

Proposition 1 (Exact transformation consistency) *Let a a finite transformation group H_a act on both the predicate and geometry of an ordered pair, and define $C_a^{\text{tr}}(x) = |H_a|^{-1} \sum_{g \in H_a} C_a(gx)$. Then $C_a^{\text{tr}}(hx) = C_a^{\text{tr}}(x)$ for every $h \in H_a$.*

Proof. For any $h \in H_a$, $C_a^{\text{tr}}(hx) = |H_a|^{-1} \sum_{g \in H_a} C_a(ghx)$. Right multiplication by h permutes the elements of the finite group, so re-indexing $g' = gh$ gives $C_a^{\text{tr}}(hx) = C_a^{\text{tr}}(x)$. The proximity and vertical-order transformations yield the identities in the main paper. For support/contact, the transformation set contains only the identity.

Family-sequence preservation. For each family a , the algorithm constructs the ordered list π_a defined in the main paper. At position j , it selects the next unselected candidate from π_{a_j} . The output relation-family sequence is therefore exactly $(a_1, \dots, a_n)$. The support/contact list follows the source ranking, so the first K positions contain exactly the same ordered support/contact subsequence as the source prefix. Therefore the support/contact selections and their order are identical at every reported K .

Optimality for each prefix. Fix a prefix length K , and let $n_a(K)$ be the number of positions assigned to family a in the source prefix. Every feasible ranking that preserves these family counts must select $n_a(K)$ candidates from each family. Because each re-ranked π_a is sorted by u_i , each estimator selects the $n_a(K)$ highest-scoring candidates in that family. Replacing one of these candidates with a lower-scoring candidate from the same family would reduce the summed score of the prefix. The support/contact subsequence is fixed, so it does not affect this comparison.

B Reproducibility and Experimental Setup

This section specifies the evaluation scope, source predictions, primary verifier, and implementation.

Feature	Definition
distance_3d	$\ c_s - c_o\ _2$
distance_xy	$\sqrt{(x_s - x_o)^2 + (y_s - y_o)^2}$
normalized_distance_3d	$2 \text{distance_3d} / (d_s^{3D} + d_o^{3D})$
normalized_distance_xy	$2 \text{distance_xy} / (d_s^{XY} + d_o^{XY})$
center_delta_z	$z_s - z_o$
normalized_center_delta_z	$2(z_s - z_o) / (h_s + h_o)$
projected_iou_xy	$I_{XY} / (A_s + A_o - I_{XY})$
projected_subject_overlap_ratio	I_{XY} / A_s
projected_object_overlap_ratio	I_{XY} / A_o
vertical_gap_subject_on_object	$b_s - t_o$
subject_bottom_z, subject_top_z	b_s, t_s
object_bottom_z, object_top_z	b_o, t_o
abs_center_delta_z	$ z_s - z_o $
abs_normalized_center_delta_z	$ 2(z_s - z_o) / (h_s + h_o) $
abs_vertical_gap_subject_on_object	$ b_s - t_o $

Table S4: The 17 predicate-independent geometric measurements in G . Projected areas use scene-coordinate XY bounds recovered from the annotation OBBs. A missing numeric value is replaced by its training-split mean before standardization.

B.1 Data, Predictors, and Evaluation Scope

Compatibility fitting uses 1,061 training scans. The 117 development scans do not overlap with the training or evaluation scans and are used for model design and selection of relation families. The shared evaluation set contains 157 official 3DSSG validation scans, 548 relation contexts, and 3,972 exact-match ground-truth relations in the support/contact, proximity, and vertical-order families. Evaluation candidates, source relation scores, and verifier-status labels are not used for fitting. The same fitted estimators are applied to all three predictors without predictor-specific refitting.

VL-SAT. We rerun the released VL-SAT Mmgnet checkpoint on the official validation scenes with its 160-object and 26-relation ontology. Evaluation disables data augmentation and uses 128 sampled points per object and 256 per union. The adapter applies a sigmoid to each relation logit and exports the scores for all 26 relation predicates for every ordered pair. This produces 957,008 candidates over 36,808 ordered pairs and 548 contexts. We retain the six predicates in the three evaluated families and join each candidate to geometry by its scan, context, subject, and object identifiers.

Open3DSG. We train Open3DSG using the official implementation and non-averaged BLIP configuration, and select the checkpoint using training and development losses only. The official preprocessing produces candidates for 533 of the 548 contexts. The evaluation retains all 548 contexts and treats the 15 contexts without candidates as empty sets.

SGFN. We evaluate the released SGFN checkpoint with its 160-object and 26-relation configuration on the same validation scenes using the same Recall denominator of 3,972 relations.

Source relation scores. The observed proximity and vertical-order score ranges are $[5.30 \times 10^{-22}, 0.9954]$ for VL-SAT, $[0.6394, 0.9281]$ for Open3DSG, and $[4.61 \times 10^{-20}, 0.5846]$ for SGFN. All evaluated scores are

positive, so multiplication by compatibility cannot reverse their order through a sign change. These ranges are observations, not calibrated probability intervals.

B.2 Primary Verifier Specification

The same fixed rule-based verifier is used for all three predictors. Proximity and vertical-order candidates use OBB-derived measurements. Support/contact candidates use reconstructed instance points and a local support score computed according to the support type. Let $\tilde{d}_{xy}$ denote normalized XY center distance, and let O_{xy} indicate that either directional projected-overlap ratio is positive. For vertical order, (δ_z^p, δ_z^o) equals $(\Delta z, \Delta z^{\text{norm}})$ for higher than and its negation for lower than. Table S5 gives the complete status boundaries.

Support/contact score. For a signed gap g , define $\rho(g; a, b) = \text{clip}(1 - (|g| - a) / (b - a), 0, 1)$. For a point count n , define $\eta(n; c) = \min(n/c, 1)$. We use the soft-support rule for the lying on predicate or when either endpoint is a pillow, cushion, blanket, clothes, or towel. Otherwise, we use the legged floor-support rule when the object is labeled as floor and the rigid-support rule in all remaining cases. We use the smallest XY-footprint expansion in $\{0, 0.05, 0.10, 0.20\}$ m that contains at least 10 support points. Here $g_{05/95}$ is the subject height at the 5th percentile minus the local support height at the 95th percentile. We define $g_{01/99}$ in the same way using the 1st and 99th percentiles. Legged floor support uses

$$s_{\text{sc}} = 0.70 \rho(g_{01/99}; 0.08, 0.18) + 0.30 \eta(n; 50),$$

with the score clipped below at 0.40 because evidence from low height percentiles alone does not declare a violation. Soft support/contact uses $s_{\text{sc}} = 0.55 s_{\text{gap}} + 0.30 \eta(n; 50) + 0.15$. For a positive gap, $s_{\text{gap}} = \rho(g_{05/95}; 0.08, 0.20)$. For a negative gap, the score is 1.0 for overlap up to 0.15 m, decreases linearly to 0.55 at 0.45 m, and remains 0.55 beyond

Family	Measurements	Satisfied	Uncertain	Violated
Proximity	$\tilde{d}_{xy}$ and directional projected overlap	O_{xy} , or $\tilde{d}_{xy} \leq 2.5$	No overlap and $2.5 < \tilde{d}_{xy} < 3.5$, or missing distance	No overlap and $\tilde{d}_{xy} \geq 3.5$
Vertical order	δ_z^p and $\tilde{\delta}_z^p$	$\delta_z^p \geq 0.25$ m and $\tilde{\delta}_z^p \geq 0.15$	Other finite values or missing measurements	$\delta_z^p \leq -0.25$ m and $\tilde{\delta}_z^p \leq -0.15$
Support/contact	Point evidence and local support score s_{sc} , computed according to support type	$s_{sc} \geq 0.70$ with the required point evidence	$0.40 \leq s_{sc} < 0.70$, missing point evidence, or unavailable rigid support plane	$s_{sc} < 0.40$ with sufficient point evidence

Table S5: Primary geometry-verifier specification. These fixed rules assign the satisfied, uncertain, and violated statuses used by verifier-derived Violation. They are evaluation rules, not training targets or learned compatibility estimates.

that value. Rigid support uses

$$s_{sc} = 0.55 \rho(g_{\text{plane}}; 0.08, 0.22) + 0.35 c_{\text{plane}} + 0.10 \eta(n; 50).$$

The local plane is the densest 0.04 m height bin within $[-0.25, 0.12]$ m of the subject bottom and within its XY footprint spanning the 5th to 95th percentiles, expanded by 0.20 m. We use this local plane only when at least 10 points are assigned to it. The value g_{plane} is the subject height at the 5th percentile minus the plane height. Its confidence is

$$c_{\text{plane}} = 0.40 \eta(n_{\text{in}}; 10) + 0.30 \min(r_{\text{in}}/0.15, 1) + 0.30 \rho(e_{\text{plane}}; 0.015, 0.060),$$

where r_{in} is the fraction of points assigned to the plane and e_{plane} is their mean absolute residual. Missing endpoint geometry is uncertain. The verifier does not assign a status to predicates outside the three evaluated families, so they are excluded from the reported Recall and Violation evaluation.

Separation from training construction. Table S2 defines how training counterfactuals are constructed, whereas Table S5 defines evaluation statuses. The two procedures share some measurements and thresholds, as made explicit in Table S3, but evaluation candidates and verifier-status labels are not used for fitting.

B.3 Implementation Details

All RelCompat3D preprocessing, fitting, and evaluation use a fixed Docker environment based on Python 3.11.9 with NumPy 1.26.4. Re-ranking requires no GPU. Predictor inference runs in separate containers. Compatibility is computed once per candidate and stored before re-ranking.

C Additional Results Supporting the Main Paper

Predicate, ordered-pair, and source-score controls test T , G , and Z . Other analyses cover components, audits, qualitative cases, and robustness.

Bold marks preferred values within each predictor and matched comparison, including ties. Captions define metric directions.

C.1 Estimator Components and Feature Removal

Component-removal controls. Tables S6–S7 isolate the linked pairwise loss and transformation averaging for both

compatibility estimators. Within each estimator, all conditions retain the same training examples, features, optimizer, source relation scores, and family-aware re-ranking procedure. The No pairwise condition removes only the linked pairwise term while retaining relation-preserving augmentation and transformation averaging. The No averaging condition uses the same fitted estimator without inference-time transformation averaging.

Predictor	Condition	$K = 50$	$K = 100$
VL-SAT	Linear full	92.77/1.97	96.58/2.95
	Linear no pairwise	92.77/1.98	96.58/2.97
	Linear no averaging	92.77/1.97	96.60/2.95
	MLP full	92.72/1.89	96.50/2.96
	MLP no pairwise	92.75/1.80	96.48/2.80
	MLP no averaging	92.75/1.89	96.45/2.96
Open3DSG	Linear full	44.18/3.42	56.85/3.24
	Linear no pairwise	44.18/3.42	56.92/3.24
	Linear no averaging	44.13/3.42	56.92/3.24
	MLP full	46.70/4.13	59.89/3.71
	MLP no pairwise	46.00/3.99	59.39/3.62
	MLP no averaging	45.07/3.98	57.35/3.68
SGFN	Linear full	74.50/2.63	93.03/3.50
	Linear no pairwise	74.50/2.64	93.03/3.53
	Linear no averaging	74.50/2.63	93.03/3.50
	MLP full	74.57/2.58	92.88/3.50
	MLP no pairwise	74.52/2.55	92.95/3.30
	MLP no averaging	74.57/2.58	93.00/3.52

Table S6: Component-removal controls. Each entry reports exact-match Recall / verifier-derived Violation percentages. Full, No pairwise, and No averaging are compared separately within each estimator.

Transformation averaging yields zero compatibility error and identical selected top- K sets under transformed representations for the full estimators and models trained without the pairwise loss. Thus transformation averaging enforces exact consistency, although its aggregate effect in Table S6 is small. Table S8 reports the feature-removal results under the same family-aware re-ranking procedure.

Every feature-removal model is refitted on the 1,061 training scans with the same linked positive-counterfactual objective, exclusion of the source relation score, and exact transformation averaging. At $K = 50$, No related measurements and Alternative evidence retain a clear Open3DSG improvement

Estimator	Condition	Family	Mean	P95	Max	Min. top- K overlap	Min. exact agreement
Linear	Full	Both	.0000	.0000	.0000	1.0000	1.0000
Linear	No pairwise	Both	.0000	.0000	.0000	1.0000	1.0000
Linear	No averaging	Proximity	.0150	.0817	.4532	.9635	.8266
Linear	No averaging	Vertical	.0014	.0080	.0192		
MLP	Full	Both	.0000	.0000	.0000	1.0000	1.0000
MLP	No pairwise	Both	.0000	.0000	.0000	1.0000	1.0000
MLP	No averaging	Proximity	.1684	.5805	.9370	.6975	.4617
MLP	No averaging	Vertical	.0402	.2896	.9157		

Table S7: Transformation checks. For each estimator and family, Mean, P95, and Max report the largest predictor-specific mean, 95th percentile, and maximum absolute compatibility differences under the applicable endpoint/predicate transformation. Top- K overlap is the Jaccard score between the selected sets before and after transformation. Exact agreement is the fraction of contexts in which those sets are identical. The last two columns report the minimum over predictors and reported K values.

Predictor	Condition	$K = 50$	$K = 100$
VL-SAT	Source	92.72/2.68	96.35/4.76
	Full	92.77/ 1.97	96.58/ 2.95
	No verifier scalar	92.80 /1.99	96.60 /3.00
	No related measurements	92.75/2.68	96.53/4.68
	Alternative evidence	92.80 /2.68	96.42/4.71
Open3DSG	Source	40.43/13.87	51.11/12.42
	Full	44.18 / 3.42	56.85 / 3.24
	No verifier scalar	43.81/ 3.42	56.87 /3.25
	No related measurements	41.19/9.54	52.64/9.55
	Alternative evidence	41.04/11.94	53.27/11.88
SGFN	Source	74.02/3.85	92.35/6.30
	Full	74.50/ 2.63	93.03 / 3.50
	No verifier scalar	74.65/2.68	93.00/3.58
	No related measurements	74.67 /3.88	92.98/6.16
	Alternative evidence	74.62/3.86	92.77/6.25

Table S8: Feature-removal analysis. Each entry reports exact-match Recall / verifier-derived Violation percentages under the same family-aware re-ranking procedure. The No verifier scalar condition removes the normalized scalar used by the verifier. The No related measurements condition removes all directly related distance or vertical measurements. Alternative evidence retains projected overlap for proximity and horizontal-distance/overlap context for vertical order.

but do not retain the full Violation reduction on VL-SAT or SGFN. At $K = 100$, all feature-removal conditions improve both reported metrics across all three predictors. These results show that performance does not depend on one verifier input alone. They do not rule out dependence on other correlated geometric measurements. The main paper reports matched controls under family-aware re-ranking.

C.2 Point- and Mesh-Based Consistency Audit

The primary verifier and compatibility estimators share some OBB-derived measurements. We therefore construct an alternative audit from reconstructed instance points and meshes. The point-based estimate uses robust distances and heights from reconstructed instance vertices. The mesh-based estimate uses area-weighted samples from reconstructed mesh

triangles. A candidate is labeled satisfied or violated only when these estimates agree. A disagreement is labeled uncertain. Thresholds are fixed from training-split positives. Both estimates exclude OBB measurements, source relation scores, learned compatibility, predictor identity, and primary verifier labels. Table S9 reports the agreement-based Violation values for both estimators at all five K values.

For each estimator, agreement-based Violation is lower than or tied with Source in all 15 predictor- K settings. The only tie is SGFN at $K = 5$. The paired 95% interval is below zero in every setting where the point estimate decreases. Point-only and mesh-only Violation values are also no higher than Source. At $K = 50$, 95.5–98.3% of selected candidates have both point and mesh measurements. Among those candidates, 71.0–85.7% receive a satisfied or violated label across predictors and estimators.

C.3 Qualitative Pair Analysis

The Open3DSG context used for the main-paper teaser also contains a complementary promotion case. The exact-label relation “desk close by chair” is verifier-satisfied, with an XY center distance of 0.436 m. RelCompat3D-Linear moves this candidate from source rank 81 to rank 30, bringing it into the top-50. Because the two candidates belong to different relation families, this promotion is not a direct exchange with the vertical-order candidate.

Figure S1 links each ordered-pair projection to its measured evidence and ranking outcome. It shows a promoted proximity candidate, a demoted vertical-order candidate, and a support/contact candidate retained in source order.

C.4 Score, Baseline, and Routing Sensitivities

Tables S10–S12 summarize the score-mapping, non-learned baseline, and matched MLP results. Table S13 then compares the family-aware rule with the matched joint P/V control.

Source-score mapping sensitivity. Monotonic mappings preserve source order but can change the product $Z_i C_i^{\text{tr},q}$. Without evaluation-based selection, we test Z^γ for $\gamma \in \{0.5, 2, 4\}$ and $\sigma(\text{logit}(\bar{Z})/\tau)$ for $\tau \in \{0.5, 2\}$, using $\bar{Z} = \text{clip}(Z, 10^{-6}, 1 - 10^{-6})$. These mappings test scale sensitivity without treating cosine scores as probabilities.

Predictor	Ranking rule	$K = 5$	$K = 10$	$K = 20$	$K = 50$	$K = 100$
VL-SAT	Source	0.93	3.61	7.27	16.43	26.00
	Linear	0.51	3.06	6.36	14.34	21.50
	MLP	0.42	2.79	6.17	14.05	21.57
Open3DSG	Source	83.20	65.18	53.80	45.96	40.87
	Linear	1.26	1.60	2.90	4.90	8.65
	MLP	16.84	13.18	9.17	8.84	12.32
SGFN	Source	0.00	2.42	5.33	11.56	21.55
	Linear	0.00	1.54	3.89	8.13	15.98
	MLP	0.00	1.32	3.55	7.63	15.88

Table S9: Point- and mesh-based consistency audit. Agreement-based Violation (% , $\downarrow$) on the shared 3DSSG validation scenes. A satisfied or violated label requires agreement between the point- and mesh-based measurements.

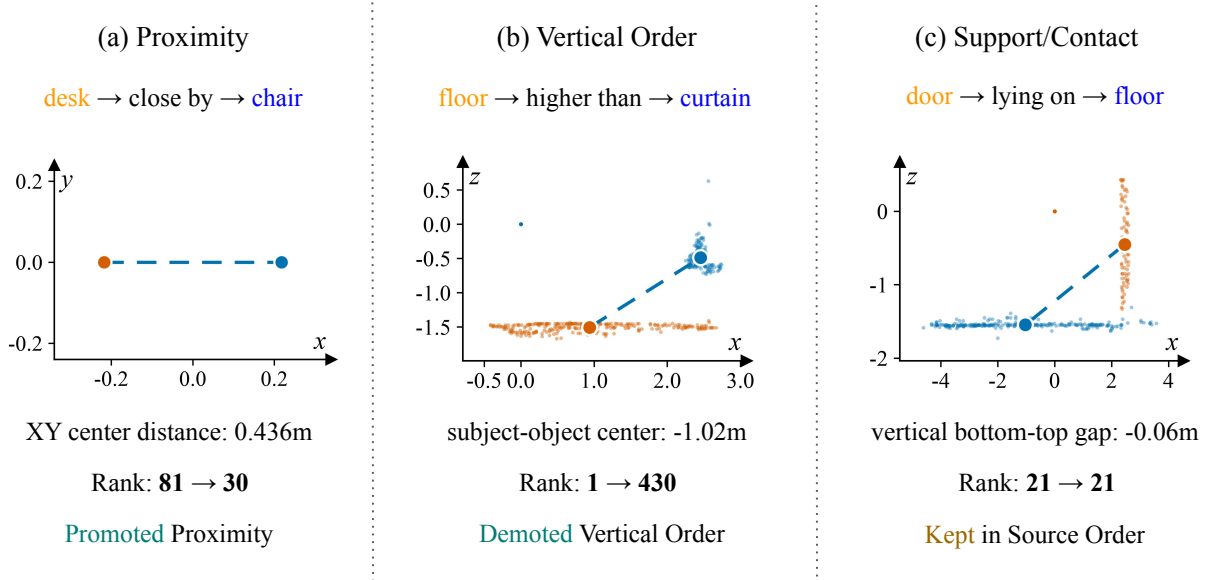


Figure S1: Qualitative ordered-pair examples. Orange marks identify the subject and blue marks identify the object. The panels show a promoted proximity candidate, a demoted vertical-order candidate, and a support/contact candidate retained in source order.

Predictor	Estimator	Five smooth mappings			Percentile mapping		
		Pass	Min. ΔR	Max. ΔV	Pass	Min. ΔR	Max. ΔV
VL-SAT	Linear	25/25	0.000	-0.036	5/5	+0.025	-0.146
	MLP	24/25	-0.025	-0.109	3/5	-0.201	-0.146
Open3DSG	Linear	25/25	+0.302	-9.163	5/5	+0.277	-8.982
	MLP	25/25	+0.277	-8.240	5/5	+0.201	-7.851
SGFN	Linear	25/25	0.000	0.000	3/5	-0.227	0.000
	MLP	25/25	0.000	0.000	3/5	-0.050	0.000

Table S10: Source-score mapping sensitivity over $K \in \{5, 10, 20, 50, 100\}$. Pass counts settings with $\Delta R \geq 0$ and $\Delta V \leq 0$ relative to Source. Minimum Recall and maximum Violation changes are percentage points. Negative ΔV is preferred. The smooth-mapping columns summarize three power mappings and two logit-temperature mappings. The percentile mapping ranks scores within each context and family.

Predictor	Ranking rule	$K = 5$	$K = 10$	$K = 20$	$K = 50$	$K = 100$
VL-SAT	Source	41.94/0.29	63.22/0.82	80.74/1.42	92.72/2.68	96.35/4.76
	Robust density	41.14/0.33	62.13/0.97	79.83/1.51	91.77/2.68	96.10/4.46
	Linear	42.07/ 0.15	63.39/0.57	80.82/1.14	92.77 /1.97	96.58 / 2.95
	MLP	42.09 / 0.15	63.47 / 0.51	80.92 / 1.09	92.72/ 1.89	96.50/2.96
Open3DSG	Source	3.42/52.05	9.87/32.89	19.89/20.99	40.43/13.87	51.11/12.42
	Robust density	4.03 /2.03	11.56/3.45	22.61/4.30	43.76/5.09	56.92/6.05
	Linear	3.73/ 0.94	11.38/ 2.33	23.62/ 3.13	44.18/ 3.42	56.85/ 3.24
	MLP	3.70/4.95	11.78 /4.97	24.67 /4.56	46.70 /4.13	59.89 /3.71
SGFN	Source	31.17 / 2.37	39.75 /3.49	49.12/3.22	74.02/3.85	92.35/6.30
	Robust density	31.14/ 2.37	39.43/3.58	48.51/3.31	73.87/4.01	91.84/6.19
	Linear	31.17 / 2.37	39.75 / 3.47	49.14/2.97	74.50/2.63	93.03 / 3.50
	MLP	31.17 / 2.37	39.75 / 3.47	49.19 / 2.96	74.57 / 2.58	92.88/ 3.50

Table S11: Comparison with the non-learned robust-density baseline on the shared 3DSSG validation scenes. Each entry reports exact-match Recall / verifier-derived Violation percentages. Linear and MLP denote the two proposed RelCompat3D estimators under the same family-aware re-ranking rule.

Predictor	Condition	R@50	V@50	R@100	V@100
Open3DSG	MLP full	46.70	4.13	59.89	3.71
	Wrong predicate	46.78	19.30	58.89	18.34
	Wrong pair	38.02	9.31	49.90	8.90
	Shuffled geometry	36.83	12.88	47.94	12.38
	Fixed-predicate swap	45.52	19.13	57.43	18.15
	Compatibility only	48.19	3.42	61.33	3.28
VL-SAT	MLP full	92.72	1.89	96.50	2.96
	Wrong predicate	90.99	5.03	94.99	8.31
	Wrong pair	91.31	2.62	95.62	4.71
	Shuffled geometry	90.26	3.00	94.71	5.49
	Fixed-predicate swap	90.99	4.99	95.09	8.13
	Compatibility only	77.84	1.61	90.18	2.05
SGFN	MLP full	74.57	2.58	92.88	3.50
	Wrong predicate	71.78	9.67	90.01	13.52
	Wrong pair	72.10	3.91	89.05	6.59
	Shuffled geometry	71.15	4.59	87.84	8.06
	Fixed-predicate swap	71.85	9.65	90.23	13.35
	Compatibility only	57.10	2.33	80.99	2.31

Table S12: Matched controls for RelCompat3D-MLP. Recall (R, $\uparrow$) and verifier-derived Violation (V, $\downarrow$) on the shared 3DSSG validation scenes. All values are percentages. Main Table 2 reports the corresponding Linear controls.

Predictor	Estimator	Re-ranking	$K = 5$	$K = 10$	$K = 20$	$K = 50$	$K = 100$
VL-SAT	Linear	Family-aware	42.07 / 0.15	63.39/0.57	80.82/1.14	92.77 /1.97	96.58/2.95
	Linear	Joint P/V	42.04/ 0.15	63.42 / 0.55	80.84 / 1.11	92.72/ 1.95	96.60 / 2.94
	MLP	Family-aware	42.09/ 0.15	63.47 /0.51	80.92/1.09	92.72/1.89	96.50/2.96
	MLP	Joint P/V	42.12 / 0.15	63.42/ 0.49	80.97 / 1.06	92.77 / 1.86	96.53 / 2.93
Open3DSG	Linear	Family-aware	3.73/ 0.94	11.38/ 2.33	23.62/ 3.13	44.18/3.42	56.85/3.24
	Linear	Joint P/V	7.98 / 0.94	15.84 / 2.33	25.33 / 3.13	44.76 / 3.39	57.83 / 3.08
	MLP	Family-aware	3.70/4.95	11.78/4.97	24.67 / 4.56	46.70 / 4.13	59.89 /3.71
	MLP	Joint P/V	7.48 / 4.17	14.85 / 4.77	23.92/4.66	43.78/4.16	56.09/ 3.68
SGFN	Linear	Family-aware	31.17 / 2.37	39.75 / 3.47	49.14/ 2.97	74.50/2.63	93.03/3.50
	Linear	Joint P/V	31.17 / 2.37	39.75 / 3.47	49.19 / 2.97	74.67 / 2.61	93.28 / 3.39
	MLP	Family-aware	31.17 / 2.37	39.75 / 3.47	49.19/ 2.96	74.57/2.58	92.88/3.50
	MLP	Joint P/V	31.17 / 2.37	39.75 / 3.47	49.27 / 2.96	75.25 / 2.57	93.13 / 3.37

Table S13: Matched routing comparison on the shared 3DSSG validation scenes. Each entry reports exact-match Recall / verifier-derived Violation percentages. Joint P/V changes only the competition between proximity and vertical-order candidates. Support/contact positions and identities remain fixed.

Table S10 summarizes all five reported K values. A setting meets both criteria when Recall is no lower and verifier-derived Violation is no higher than Source. Every Linear setting and all but one MLP setting meet both criteria under the smooth mappings. In the MLP exception, VL-SAT at $K = 50$ under Z^4 , the paired Recall interval includes zero and Violation decreases. The percentile mapping produces small Recall losses in some settings without increasing Violation. Across all tested mappings, the lowest pair-weighted Kendall rank correlation is 0.810 and the lowest top-50 set overlap (Jaccard score) is 0.815. These results show limited sensitivity over the tested mappings, not score-scale invariance.

Robust-density baseline. The non-learned robust-density baseline uses only geometric measurements from training-split positive relations. It computes predicate-specific medians and interquartile ranges and uses transformation averaging, the same product score, and the same family-aware re-ranking rule. It does not use verifier labels from the evaluation or constructed counterfactual negatives. Table S11 reports all K values. At $K = 50$, both learned estimators have higher Recall and lower Violation than this baseline for all three predictors. The baseline improves both metrics over a learned estimator only for MLP on Open3DSG at $K = 5$. The remaining comparisons favor a learned estimator on both metrics or show a trade-off.

Compatibility estimators and matched fusion. The MLP is a matched nonlinear estimator rather than a separate method. RankAvg and reciprocal rank fusion (RRF) use the Linear compatibility ranks:

$$\begin{aligned} \text{RankAvg}_i &= \frac{1}{2} (\tilde{r}_i^Z + \tilde{r}_i^C), \\ \text{RRF}_i &= \frac{1}{60 + r_i^Z} + \frac{1}{60 + r_i^C}. \end{aligned} \quad (\text{S1})$$

Here r_i^Z and r_i^C are within-family ranks, and tildes denote normalized ranks. Main Table 1 reports both fusion rules at every K . Their predictor-dependent Recall–Violation trade-offs show that neither RankAvg nor RRF consistently improves both metrics over the product score across predictors and reported K values.

Matched MLP controls. Table S12 reports the matched controls for the MLP estimator. Main Table 2 reports the corresponding Linear controls. Across both reported K values and both compatibility estimators, using incorrect predicates or mismatched ordered-pair measurements increases verifier-derived Violation. Compatibility-only ordering remains close to the full estimator on Open3DSG but loses substantial Recall on VL-SAT and SGFN. These controls support the predictor-dependent role of the source relation score described in the main paper.

Matched routing controls. The *family-aware* rule preserves the source relation-family label at every rank position. It re-ranks proximity and vertical-order candidates in separate family lists. The matched joint proximity/vertical control (Joint P/V) uses the same candidates, compatibility estimates, product scores, and support/contact positions and

identities. It differs only by placing proximity and vertical-order candidates in one list. Table S13 reports this comparison at all five K values.

No rule is best in every setting. For Open3DSG, Joint P/V raises Linear Recall at all five K values. For the MLP, it raises Recall at $K = 5, 10$ but lowers it at $K = 50, 100$. The paired Recall intervals for both decreases are below zero. At Open3DSG $K = 50$, the MLP control changes the selected proximity/vertical-order counts from 6,295/6,508 to 3,423/9,380. All 13,833 support/contact selections remain unchanged. We therefore interpret family-aware re-ranking as a constraint that preserves relation-family composition, not as a rule that maximizes aggregate performance.

C.5 Paired Scan-Level Intervals

Paired 95% intervals use 1,000 scan-level bootstrap resamples. All contexts from a sampled scan are resampled together, and the same resample indices are used for Source and each RelCompat3D variant. Table S14 reports the paired $K = 50$ changes summarized in the main paper.

Estimator	Predictor	$\Delta R@50$	$\Delta V@50$
Linear		+0.05	−0.70
	VL-SAT	[−0.05, +0.18]	[−0.88, −0.55]
		+3.75	−10.45
	Open3DSG	[+2.96, +4.50]	[−11.06, −9.86]
MLP		+0.48	−1.22
	SGFN	[+0.29, +0.69]	[−1.44, −1.01]
		0.00	−0.79
	VL-SAT	[−0.10, +0.10]	[−0.98, −0.62]
MLP		+6.27	−9.74
	Open3DSG	[+5.02, +7.49]	[−10.31, −9.13]
		+0.55	−1.27
	SGFN	[+0.32, +0.81]	[−1.51, −1.06]

Table S14: Paired scan-level changes at $K = 50$. Each entry reports the RelCompat3D-minus-Source change and its paired 95% bootstrap confidence interval in percentage points. Positive Recall and negative Violation changes are favorable.

For both estimators on Open3DSG and SGFN, the Recall and Violation intervals exclude zero in the favorable direction. For VL-SAT, the Recall intervals contain zero while the Violation intervals remain below zero.

C.6 Relation-Family Composition

Table S15 reports selected counts, shares, and family-specific metric changes within the selected top-100 predictions. Deltas are RelCompat3D-Linear minus Source.

Predictor	Family	Selected	Share	ΔR	ΔV
VL-SAT	Support/contact	24,021	43.8%	0.00	0.00
VL-SAT	Proximity	16,513	30.1%	+0.40	-0.76
VL-SAT	Vertical order	14,266	26.0%	+0.51	-6.09
Open3DSG	Support/contact	27,162	51.8%	0.00	0.00
Open3DSG	Proximity	10,929	20.8%	+9.68	-4.43
Open3DSG	Vertical order	14,361	27.4%	+15.38	-30.14
SGFN	Support/contact	23,616	43.1%	0.00	0.00
SGFN	Proximity	9,562	17.4%	+1.42	-1.50
SGFN	Vertical order	21,622	39.5%	+0.51	-6.43

Table S15: Family composition and family-specific metrics within the selected top-100 predictions. Counts and shares are identical for Source, RelCompat3D-Linear, and RelCompat3D-MLP because family-aware re-ranking preserves the source relation-family label sequence. Deltas are RelCompat3D-Linear minus Source in percentage points, and positive ΔV is a regression.

Support/contact is exactly unchanged because the ranking procedure preserves its selected subsequence and the source relation-family label sequence at every K . Vertical order drives most of the aggregate reduction. The result is consequently a scoped improvement, not a uniform family claim.

C.7 Sensitivity to Verifier Uncertainty

The reported Violation@ K includes uncertain candidates in its denominator but not its numerator. We also compute decidable-only Violation over satisfied and violated candidates, the uncertainty rate, and an uncertain-as-violation variant that counts every uncertain candidate as a violation. Table S16 gives the $K = 100$ comparison.

Predictor	Ranking rule	Dec. cov.	V_{all}	V_{dec}	U	$V_{u \rightarrow v}$
VL-SAT	Source	65.36	4.76	7.29	34.64	39.41
VL-SAT	Linear	65.27	2.95	4.52	34.73	37.68
VL-SAT	MLP	65.42	2.96	4.53	34.58	37.55
Open3DSG	Source	58.39	12.42	21.26	41.61	54.02
Open3DSG	Linear	59.15	3.24	5.48	40.85	44.09
Open3DSG	MLP	57.35	3.71	6.48	42.65	46.37
SGFN	Source	62.68	6.30	10.05	37.32	43.62
SGFN	Linear	61.97	3.50	5.65	38.03	41.53
SGFN	MLP	62.19	3.50	5.63	37.81	41.31

Table S16: Primary-verifier status coverage and uncertainty sensitivity at $K = 100$. Every selected candidate receives a verifier status, so status coverage is 100%. All entries are percentages. Decidable coverage is the fraction labeled satisfied or violated. V_{all} is the reported Violation, and V_{dec} is decidable-only Violation. U is the uncertain fraction. $V_{u \rightarrow v}$ counts uncertain candidates as violations. Lower is better for V and U , while higher is better for coverage.

For RelCompat3D-Linear, decidable-only and uncertain-as-violation V are lower than Source for all three predictors, and the paired decidable-only intervals exclude zero. Uncertainty does not decrease uniformly. Across both estimators, all three Violation variants are no higher than Source in all

30 predictor- K settings. These results show that the Violation reduction does not depend on one treatment of uncertain candidates.

Additional robustness checks. These checks do not change the main comparisons or broaden the claim beyond the shared 3DSSG validation scenes. We repeat training with five pre-specified seeds. These repetitions assess variability and do not select the reported model. Linear results are identical across the five fits. For MLP, Recall is no lower and Violation is no higher than Source in every predictor- K -seed setting except one VL-SAT fit at $K = 50$. That fit retrieves one fewer exact relation while retaining lower Violation.

We vary four settings one at a time. For negatives per positive and the pairwise weight, we test $\{1, 4\}$ and $\{0.125, 0.5\}$, compared with 2 and 0.25. For the proximity-negative threshold and vertical absolute margin, we test $\{2.0, 3.0\}$ and $\{0.20, 0.30\}$ m, compared with 2.5 and 0.25 m. Across these eight conditions, Linear changes by at most 0.35 Recall and 0.20 Violation percentage points without producing lower Recall or higher Violation than Source.